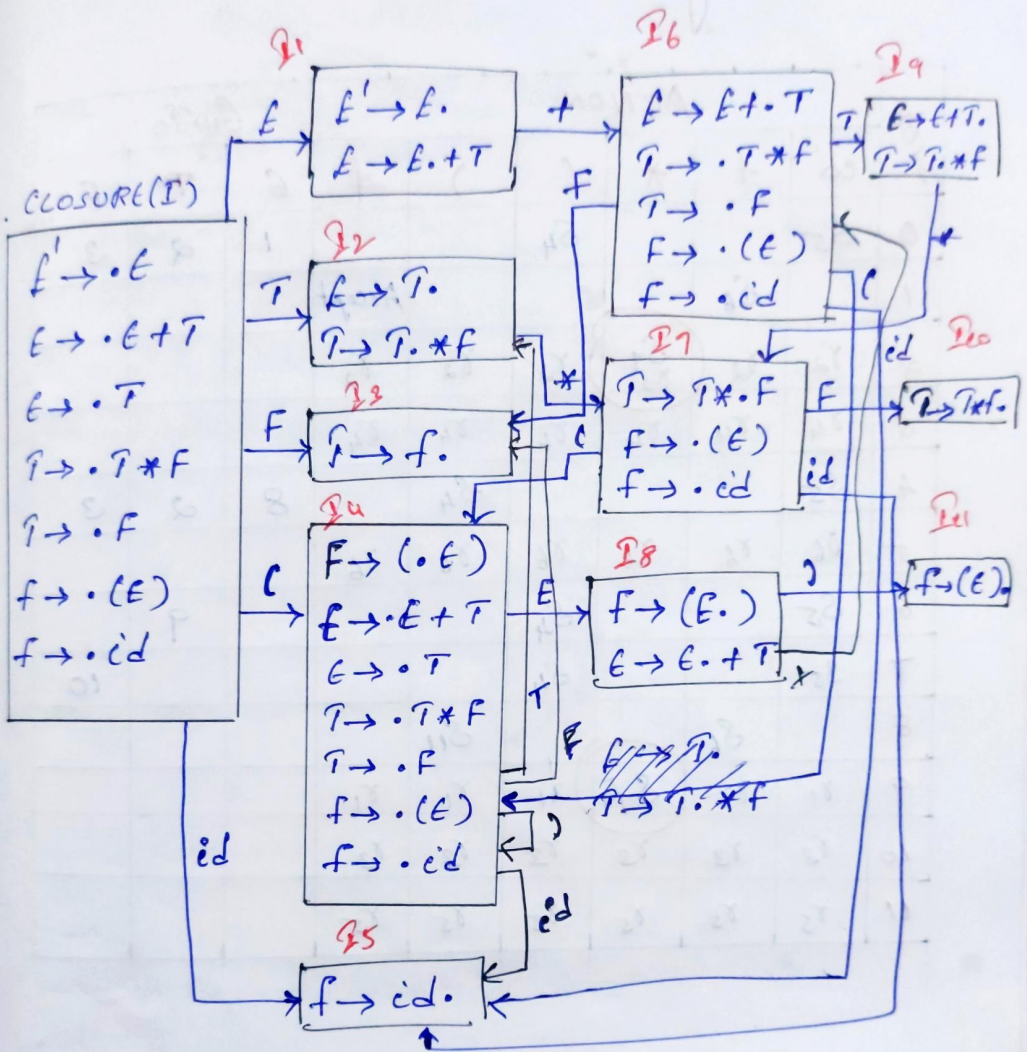


Compute LR(0) Automaton



1. $E \rightarrow E + T$
2. $E \rightarrow T$
3. $T \rightarrow T * F$
4. $T \rightarrow F$
5. $F \rightarrow (E)$
6. $F \rightarrow id$

LR(0) Parsing Table

STATE	ACTION						GoTo		
	cd	+	*	(	)	\$	G	T	F
0	S ₅			S ₄			1	2	3
1		S ₆				Accept			
2	r ₂	r ₂	S ₇ r ₂	r ₂	r ₂	r ₂			
3	r ₄	r ₄	r ₄	r ₄	r ₄	r ₄			
4	S ₅				S ₄		8	2	3
5	r ₆	r ₆	r ₆	r ₆	r ₆	r ₆			
6	S ₅			S ₄				9	
7	S ₅			S ₄					10
8		S ₆			S ₁₁				
9	r ₁	r ₁	S ₇ r ₁	r ₁	r ₁	r ₁			
10	r ₃	r ₃	r ₃	r ₃	r ₃	r ₃			
11	r ₅	r ₅	r ₅	r ₅	r ₅	r ₅			

→ State 2 and 9, there is a conflict known as shift reduce conflict.

→ Parser get Confuse whether to perform shift or reduce

→ Since in LR(0), we implement reduce to the entire row, two conflicts occurs (shift reduce conflict and reduce, reduce conflict)

→ LR(0) least powerful conflict